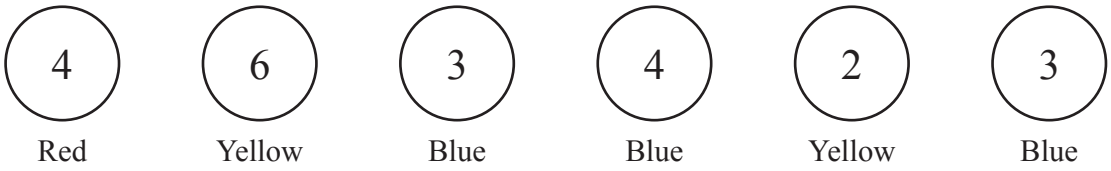
0580_w20_qp_42_Q6

6



The diagram shows six discs.
Each disc has a colour and a number.

(a) One disc is picked at random.

Write down the probability that

(i) the disc has the number 4,

..... [1]

(ii) the disc is red and has the number 3,

..... [1]

(iii) the disc is blue and has the number 4.

..... [1]

(b) Two of the six discs are picked at random **without** replacement.

Find the probability that

(i) both discs have the number 3,

..... [2]

(ii) both discs have the same colour.

..... [3]

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(c) Two of the six discs are picked at random **with** replacement.

Find the probability that both discs have the same colour.

..... [3]

0580_w20_ms_42_Q6

Question	Answer	Marks	Partial Marks
6(a)(i)	$\frac{1}{3}$ oe	**1**	
6(a)(ii)	0	**1**	
6(a)(iii)	$\frac{1}{6}$ oe	**1**	
6(b)(i)	$\frac{1}{15}$ oe	**2**	**M1** for $\frac{2}{6}\times\frac{1}{5}$ or equivalent method
6(b)(ii)	$\frac{4}{15}$ oe	**3**	**M2** for $\frac{2}{6}\times\frac{1}{5}+\frac{3}{6}\times\frac{2}{5}$ or equivalent method or **M1** for $\frac{2}{6}\times\frac{1}{5}$ oe seen or $\frac{3}{6}\times\frac{2}{5}$ oe seen
6(c)	$\frac{7}{18}$ oe	**3**	**M2** for $\left(\frac{1}{6}\right)^2+\left(\frac{2}{6}\right)^2+\left(\frac{3}{6}\right)^2$ oe or **M1** for one correct product seen or sample space with 14 correct pairs identified